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Electrical connector with improved grounding piece

Abstract

An electric connector includes an insulating body, a first terminal module, a second terminal module and a grounding piece. The first terminal module includes a number of first conductive terminals each having a first tail portion. The second terminal module includes a number of second conductive terminals each having a second tail portion. The ground piece includes a first grounding tab and a second grounding tab. The first conductive terminals include a first ground terminal. The second conductive terminals include a second ground terminal. The first grounding tab is stacked on and in contact with the first tail portion of the first ground terminal. The second grounding tab is stacked on and in contact with the second tail portion of the second ground terminal. Such an arrangement simplifies the structure of the grounding piece, and can also realize a shielding function.

Inventors: Fan; Min (Shenzhen, CN), Pu; Yingbin (Shenzhen, CN), Huang; Cunting (Shenzhen, CN)

Applicant: Luxshare Precision Industry Company Limited (Shenzhen, CN)

Family ID: 84719904

Assignee: LUXSHARE PRECISION INDUSTRY COMPANY LIMITED (Shenzhen, CN)

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Primary Examiner: Jimenez; Oscar C

Attorney, Agent or Firm: Birch, Stewart, Kolasch & Birch, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This patent application claims priority of a Chinese Patent Application No. 202211360617.5,

filed on Nov. 2, 2022 and titled “ELECTRICAL CONNECTOR”, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to an electrical connector, which belongs to a technical field of connectors.

BACKGROUND

(3) With the increasing requirements of the electrical connector for data transmission quality, how to reduce the interference problem of conductive terminals in the process of data transmission is a technical problem to be solved by those of ordinary skill in the art. In order to solve the above problem, the electrical connector in the relevant technology is also provided with a grounding piece which has a plurality of ground terminals. However, this design leads to a more complicated structure of the grounding piece and is not easy to manufacture.

SUMMARY

(4) An object of the present disclosure is to provide an electrical connector with simple structure and shielding performance.

(5) In order to achieve the above object, the present disclosure adopts the following technical solution: an electrical connector, including: an insulating body, the insulating body including a mating surface, a mounting surface opposite the mating surface, a mating slot extending through the mating surface, and a mounting space extending through the mounting surface; the mounting space being in communication with the mating slot; a first terminal module, the first terminal module including a first insulating block and a plurality of first conductive terminals, each first conductive terminal including a first fixing portion fixed to the first insulating block, a first resilient arm extending from the first fixing portion, and a first tail portion; the first resilient arm being provided with a first contact portion extending into the mating slot; a second terminal module, the second terminal module including a second insulating block and a plurality of second conductive terminals, each second conductive terminal including a second fixing portion fixed to the second insulating block, a second resilient arm extending from the second fixing portion, and a second tail portion, the second resilient arm being provided with a second contact portion extending into the mating slot, the first contact portions of the first conductive terminals and the second contact portions of the second conductive terminals being located on opposite sides of the mating slot, respectively; and a grounding piece, the grounding piece including a body portion and a plurality of grounding tabs connected to the body portion, the plurality of grounding tabs including a first grounding tab and a second grounding tab; wherein the first tail portions of the plurality of first conductive terminals and the second tail portions of the plurality of second conductive terminals are located on a same side and arranged in a row in a first direction; the plurality of first conductive terminals include a first ground terminal; the plurality of second conductive terminals include a second ground terminal; the first grounding tab is stacked at least partially on the first tail portion of the first ground terminal and is in contact with the first tail portion of the first ground terminal; and the second grounding tab is stacked at least partially on the second tail portion of the second ground terminal and is in contact with the second tail portion of the second ground terminal.

(6) In order to achieve the above object, the present disclosure adopts the following technical solution: an electrical connector, including: an insulating body including a mating surface and a mating slot extending through the mating surface; a first terminal module including a plurality of first conductive terminals, each first conductive terminal including a first resilient arm and a first tail portion, the first resilient arm being provided with a first contact portion extending into the mating slot; a second terminal module including a plurality of second conductive terminals, each second conductive terminal including a second resilient arm and a second tail portion, the second resilient arm being provided with a second contact portion extending into the mating slot, the first contact portions of the first conductive terminals and the second contact portions of the second

conductive terminals being located on opposite sides of the mating slot, respectively; a grounding piece including a body portion and a plurality of grounding tabs connected to the body portion, the plurality of grounding tabs including a first grounding tab and a second grounding tab; and a plurality of cables, part of the plurality of cables being connected to part of the first tail portions of the first conductive terminals, and another part of the plurality of cables are connected to part of the second tail portions of the second conductive terminals; the plurality of cables including a ground cable connected to the body portion; wherein the first tail portions of the plurality of first conductive terminals and the second tail portions of the plurality of second conductive terminals are located on a same side and arranged in a row in a first direction; the plurality of first conductive terminals include a first ground terminal; the plurality of second conductive terminals include a second ground terminal; the first grounding tab is stacked at least partially on the first tail portion of the first ground terminal and is in contact with the first tail portion of the first ground terminal; and the second grounding tab is stacked at least partially on the second tail portion of the second ground terminal and is in contact with the second tail portion of the second ground terminal.

(7) Compared to the prior art, the present disclosure includes the plurality of first conductive terminals having the first ground terminal and the plurality of second conductive terminals having the second ground terminal. The grounding piece includes the first grounding tab and the second grounding tab. By making the first grounding tab and the second grounding tab contact the first tail portion of the first ground terminal and the second tail portion of the second ground terminal, respectively, it simplifies the structure of the grounding piece and also enables a shielding function.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a perspective schematic view of an electrical connector in accordance with an embodiment of the present disclosure;
- (2) FIG. 2 is a perspective schematic view of FIG. 1 at another angle;
- (3) FIG. 3 is a partially perspective exploded view of FIG. 1;
- (4) FIG. 4 is a partially perspective exploded view of FIG. 3 at another angle;
- (5) FIG. 5 is a partially enlarged view of a circled portion B of FIG. 4;
- (6) FIG. 6 is a perspective exploded view of a terminal module and an insulating body in FIG. 4;
- (7) FIG. 7 is a perspective exploded view of FIG. 6 at another angle;
- (8) FIG. 8 is a top view of the terminal module in FIG. 6;
- (9) FIG. 9 is a partially exploded perspective view of the terminal module in FIG. 7;
- (10) FIG. 10 is a partially perspective exploded view of FIG. 9 at another angle;
- (11) FIG. 11 is a partially perspective exploded view of FIG. 10 at another angle;
- (12) FIG. 12 is a further partially perspective exploded view of FIG. 10;
- (13) FIG. 13 is a partially perspective exploded view of FIG. 12 at another angle;
- (14) FIG. 14 is a schematic cross-sectional view taken along line C-C in FIG. 6;
- (15) FIG. 15 is a partially enlarged view of a circled portion D of FIG. 10;
- (16) FIG. 16 is a partially perspective schematic view of the electrical connector of the present disclosure when assembled with cables; and
- (17) FIG. 17 is a partially enlarged view of a circled portion E in FIG. 16.

DETAILED DESCRIPTION

(18) Exemplary embodiments will be described in detail here, examples of which are shown in drawings. When referring to the drawings below, unless otherwise indicated, same numerals in different drawings represent the same or similar elements. The examples described in the following exemplary embodiments do not represent all embodiments consistent with this application. Rather, they are merely examples of devices and methods consistent with some aspects of the application

as detailed in the appended claims.

(19) The terminology used in this application is only for the purpose of describing particular embodiments, and is not intended to limit this application. The singular forms “a”, “said”, and “the” used in this application and the appended claims are also intended to include plural forms unless the context clearly indicates other meanings.

(20) It should be understood that the terms “first”, “second” and similar words used in the specification and claims of this application do not represent any order, quantity or importance, but are only used to distinguish different components. Similarly, “an” or “a” and other similar words do not mean a quantity limit, but mean that there is at least one; “multiple” or “a plurality of” means two or more than two. Unless otherwise noted, “front”, “rear”, “lower” and/or “upper” and similar words are for ease of description only and are not limited to one location or one spatial orientation. Similar words such as “include” or “comprise” mean that elements or objects appear before “include” or “comprise” cover elements or objects listed after “include” or “comprise” and their equivalents, and do not exclude other elements or objects. The term “a plurality of” mentioned in the present disclosure includes two or more.

(21) Hereinafter, some embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. In the case of no conflict, the following embodiments and features in the embodiments can be combined with each other.

(22) Referring to FIGS. 1 to 7, the present disclosure discloses an electrical connector **100** including an insulating body **1**, a terminal module **2** at least partially mounted to the insulating body **1**, and a metal shell **3** wrapped around the insulating body **1**. In the embodiment illustrated in the present disclosure, the electrical connector **100** is an HDMI cable connector. The electrical connector **100** is configured to connect a plurality of cables **200**. Of course, it is understandable to those skilled in the art that in other embodiments, the electrical connector **100** can also be other types of electrical connectors.

(23) Referring to FIGS. 1 to 4, in the embodiment illustrated in the present disclosure, the metal shell **3** includes a first metal shell **3a** and a second metal shell **3b** mating with the first metal shell **3a**. The first metal shell **3a** and the second metal shell **3b** are sleeved on the insulating body **1**. The first metal shell **3a** is of a one-piece configuration and made of a metal material. The first metal shell **3a** includes a flange portion **31**, a sleeve portion **32** extending forwardly from the flange portion **31**, and a holding portion **33** extending backwardly from the flange portion **31**. In the embodiment illustrated in the present disclosure, the sleeve portion **32** has a shape and dimensions conforming to HDMI standards. The flange portion **31** protrudes peripherally from the sleeve portion **32** so as to be able to prevent the sleeve portion **32** from being excessively inserted into a mating connector (not shown). The holding portion **33** has a top wall **331**, a bottom wall **332** opposite to the top wall **331**, a first side wall portion **333** connecting the top wall **331** and the bottom wall **332** on one side, and a second side wall portion **334** connecting the top wall **331** and the bottom wall **332** on the other side. The bottom wall **332** is provided with a first locking hole **3321**. The top wall **331** is provided with a second locking hole **3311**. Both the first locking hole **3321** and the second locking hole **3311** are configured to lock with the terminal module **2** so as to prevent the terminal module **2** from detaching from the first metal shell **3a**.

(24) The second metal shell **3b** includes a sleeve section **34**. The sleeve section **34** is sleeved to the holding portion **33** of the first metal shell **3a**.

(25) Referring to FIGS. 1, 6 and 7, the insulating body **1** includes a mating surface **11**, a mounting surface **12** opposite to the mating surface **11**, a mating slot **110** extending forwardly through the mating surface **11**, and a mounting space **120** extending backwardly through the mounting surface **12**. The mounting space **120** is in communication with the mating slot **110**. The terminal module **2** is mounted forwardly from the mounting surface **12** into the mounting space **120**. Besides, the insulating body **1** is provided with a plurality of first terminal receiving slots **13** in communication with the mating slot **110** and a plurality of second terminal receiving slots **14** in communication

with the mating slot **110**. The first terminal receiving slots **13** and the second terminal receiving slots **14** are located on opposite sides (e.g., a lower side and an upper side) of the mating slot **110** along a thickness direction of the insulation body **1**. The first terminal receiving slots **13** and the second terminal receiving slots **14** are staggered along the thickness direction of the insulating body **1**. In the embodiment illustrated in the present disclosure, the first terminal receiving slots **13** and the second terminal receiving slots **14** are provided further extending forwardly through the mating surface **11**. As a result, the first terminal receiving slots **13** and the second terminal receiving slots **14** can be designed longer so as to provide better heat dissipation to the conductive terminals of the terminal module **2**.

(26) Referring to FIGS. **6** and **7**, in the embodiment illustrated in the present disclosure, the insulating body **1** is further provided with a locking opening **15** for locking with the terminal module **2**.

(27) Referring to FIGS. **6** to **14**, the terminal module **2** includes a first terminal module **4**, a second terminal module **5**, and a grounding piece **6**.

(28) The first terminal module **4** includes a first insulating block **41** and a plurality of first conductive terminals **42** fixed to the first insulating block **41**. In the illustrated embodiment of the present disclosure, the plurality of first conductive terminals **42** are insert-molded with the first insulating block **41**. Of course, in other embodiments, the plurality of first conductive terminals **42** may also be fixed to the first insulating block **41** by assembling.

(29) Similarly, the second terminal module **5** includes a second insulating block **51** and a plurality of second conductive terminals **52** fixed to the second insulating block **51**. In the illustrated embodiment of the present disclosure, the plurality of second conductive terminals **52** are insert-molded with the second insulating block **51**. Of course, in other embodiments, the plurality of second conductive terminals **52** may also be fixed to the second insulating block **51** by assembling.

(30) The first insulating block **41** includes a first base portion **411** and a first protruding portion **412** extending forwardly from the first base portion **411**. The first protruding portion **412** is provided with a plurality of positioning posts **413**. The first base portion **411** is provided with a boss **414**. The boss **414** includes an assembly surface **4141**, a plurality of receiving grooves **4142** recessed downwardly from the assembly surface **4141**, and a recess **4143** recessed downwardly from the assembly surface **4141**. The recess **4143** is located at the outermost side of the plurality of receiving grooves **4142**.

(31) The first base portion **411** includes a first locking protrusion **4110** locked in the first locking hole **3321**. In addition, the first base portion **411** also includes a first side wall surface **4111**, a second side wall surface **4112** opposite to the first side wall surface **4111**, a first installation groove **4113** recessed downwardly from the assembly surface **4141** and partly extending through the first side wall surface **4111** to one side, and a second installation groove **4114** recessed downwardly from the assembly surface **4141** and partly extending through the second side wall surface **4112** to the other side.

(32) Each first conductive terminal **42** is provided with a first fixing portion **423** fixed in the first insulating block **41**, a first resilient arm **421** extending forwardly from a front end of the first fixing portion **423**, a first bent portion **424** bent upwardly from a rear end of the first fixing portion **423**, and a first tail portion **422** extending backwardly from the first bent portion **424**. The first resilient arm **421** extends forwardly beyond the first insulating block **41**. The first resilient arm **421** is cantilevered and extends into the mating slot **110**. The first resilient arm **421** is provided with a first contact portion **4211** for electrically mating with the mating connector. A plane in which the first tail portion **422** is located is higher than a plane in which the first fixing portion **423** is located. This design facilitates better exposure of the first tail portions **422** to the assembly surface **4141**.

(33) The second insulating block **51** includes a second base portion **511** and a second protruding portion **512** extending forwardly from the second base portion **511**. The second protruding portion **512** is provided with a plurality of positioning holes **513** for receiving the positioning posts **413**.

The second base portion **511** includes a second locking protrusion **5110** locked in the second locking hole **3311**. The second locking protrusion **512** further includes a locking projection **5121** locked in the locking opening **15** of the insulating body **1**.

(34) Each second conductive terminal **52** is provided with a second fixing portion **523** fixed in the second insulating block **51**, a second resilient arm **521** extending forwardly from a front end of the second fixing portion **523**, a second bent portion **524** bent downwardly from a rear end of the second fixing portion **523**, and a second tail portion **522** extending backwardly from the second bent portion **524**. The second resilient arm **521** extends beyond the second insulating block **51**. The second resilient arm **521** is cantilevered and extends into the mating slot **110**. The second resilient arm **521** is provided with a second contact portion **5211** for electrically mating with the mating connector. A plane in which the second tail portion **522** is located is lower than a plane in which the second fixing portion **523** is located. This design facilitates better installation of the second tail portions **522** in the receiving grooves **4142** of the first insulating block **41**.

(35) The first contact portions **4211** and the second contact portions **5211** extend toward the middle. That is, the first contact portions **4211** protrude toward the second contact portions **5211**, and the second contact portions **5211** protrude toward the first contact portions **4211**. With such arrangement, the first contact portions **4211** and the second contact portions **5211** can jointly clamp a tongue plate of the mating connector (not shown). The second tail portions **522** extend backwardly beyond the second insulating block **51**. In the embodiment illustrated in the present disclosure, each of the first tail portions **422** of the first conductive terminals **42** and the second tail portions **522** of the second conductive terminals **52** is in a shape of a flat plate. The first tail portions **422** of the first conductive terminals **42** and the second tail portions **522** of the second conductive terminals **52** are located on the same side. In the embodiment illustrated in the present disclosure, the first tail portions **422** of the first conductive terminals **42** and the second tail portions **522** of the second conductive terminals **52** are exposed to the assembly surface **4141** so as to facilitate fixing to corresponding cables **200** by soldering or welding.

(36) The first tail portions **422** of the first conductive terminals **42** and the second tail portions **522** of the second conductive terminals **52** are arranged in a row along a first direction **A1**.

(37) Referring to FIG. **8** and FIG. **12**, in the embodiment illustrated in the present disclosure, the first conductive terminals **42** include a plurality of first signal terminals and a plurality of first ground terminals **G1**. The second conductive terminals **52** include a plurality of second signal terminals and a plurality of second ground terminals **G2**.

(38) Referring to FIGS. **10** and **15**, in the embodiment illustrated in the present disclosure, the grounding piece **6** is made of a metal material and is of a one-piece configuration. The grounding piece **6** is provided with a body portion **61**, a plurality of grounding tabs **62** connected to the body portion **61**, a first spring arm **631** extending forwardly from one side of the body portion **61**, and a second spring arm **632** extending forwardly from the other side of the body portion **61**. The first spring arm **631** is provided with a first curved protrusion **6311** protruding away from the second spring arm **632**. The second spring arm **632** is provided with a second curved protrusion **6311** protruding away from the first spring arm **631**. The first spring arm **631** and the second spring arm **632** are held in the first installation slot **4113** and the second installation slot **4114** of the first insulating block **41**, respectively. The first curved protrusion **6311** and the second curved protrusion **6321** protrude beyond the first side wall surface **4111** and the second side wall surface **4112**, respectively. The first curved protrusion **6311** and the second curved protrusion **6321** are abutted against the metal shell **3**. In the embodiment illustrated in the present disclosure, the first curved protrusion **6311** and the second curved protrusion **6321** are abutted against the first sidewall portion **333** and the second sidewall portion **334** of the first metal shell **3a**, respectively.

(39) The grounding piece **6** includes a first fixing tab **6312** extending downwardly from the first spring arm **631** and a second fixing tab **6322** extending downwardly from the second spring arm **632**. The first fixing tab **6312** and the second fixing tab **6322** are inserted in and fixed to the first

insulating block **41**. The body portion **61** is abutted against a rear end of the boss **414**.

(40) The grounding piece **6** further includes a plurality of third fixing tabs **611** bent from the body portion **61**. The third fixing tabs **611** are inserted in and fixed to the first insulating block **41**. Preferably, the first fixing tab **6312**, the second fixing tab **6322** and the third fixing tabs **611** are provided with a plurality of barbs so as to improve the fixing effect with the first insulating block **41**.

(41) The plurality of grounding tabs **62** include a plurality of first grounding tabs **621** and a plurality of second grounding tabs **622**. The grounding piece **6** further includes a back-up grounding tab **623** extending forwardly from the body portion **61**. The back-up grounding tab **623** is received in the recess **4143** of the first insulating block **41**. When the cables **200** are provided with a grounding cable **208** corresponding to the back-up grounding tab **623** (as shown in FIG. 17), the grounding cable **208** is in contact with the back-up grounding tab **623**, for example through soldering or welding. When the cables **200** are not provided with a grounding cable corresponding to the back-up grounding tab **623**, the back-up grounding tab **623** is not directly connected to the cables **200**.

(42) In the embodiment illustrated in the present disclosure, when the grounding piece **6** is fixed to the first insulating block **41**, the first grounding tabs **621** are stacked at least partially on and in contact with the first tail portions **422** of the first ground terminals **G1**, and the second grounding tabs **622** are stacked at least partially on and in contact with the second tail portions **522** of the second ground terminals **G2**. With such arrangement, the grounding piece **6**, the first ground terminals **G1**, the second ground terminals **G2** and the metal shell **3** are connected together, thereby increasing the grounding area and improving the shielding effect.

(43) Referring to FIG. 15, in the embodiment illustrated in the present disclosure, the first grounding tab **621** includes a first root portion **6211** connected to the body portion **61**, a first inclined portion **6212** connected to the first root portion **6211**, and a first extension portion **6213** connected to the first inclined portion **6212**. The first extension portion **6213** is in contact with a corresponding first tail portion **422** of the first ground terminal **G1**. Specifically, the first tail portion **422** of the first ground terminal **G1** includes a first plane **4221**. The first extension portion **6213** includes a first abutting plane **6214** in contact with the first plane **4221**. Preferably, the first grounding tab **621** is fixed to the first tail portion **422** of the first ground terminal **G1** by soldering or welding.

(44) The second grounding tab **622** includes a second root portion **6221** connected to the body portion **61**, a second inclined portion **6222** connected to the second root portion **6221**, and a second extension portion **6223** connected to the second inclined portion **6222**. The second extension portion **6223** is in contact with a corresponding second tail portion **522** of the second ground terminal **G2**. Specifically, the second tail portion **522** of the second ground terminal **G2** includes a second plane **5221**. The second grounding tab **622** includes a second abutting plane **6224** in contact with the second plane **5221**. Preferably, the second grounding tab **622** is fixed to the second tail portion **522** of the second ground terminal **G2** by soldering or welding.

(45) The first insulating block **41** and the second insulating block **51** are provided with mating buckling structures. In the illustrated embodiment of the present disclosure, the buckling structures include but not limited to the positioning posts **413** and the positioning holes **513**.

(46) In an embodiment of the present disclosure, the first tail portion **422** of the first signal terminal is configured to be soldered or welded to a first signal cable **201**; and the second tail portion **522** of the second signal terminal is configured to be soldered or welded to a second signal cable **202**. The first tail portion **422** of the first ground terminal **G1** and the second tail portion **522** of the second ground terminal **G2** are configured to be soldered or welded to a ground cable **203**. The cables **200** include the first signal cable **201**, the second signal cable **202** and the ground cable **203**.

(47) Compared to the prior art, the present disclosure includes the plurality of first conductive terminals **42** having the first ground terminal **G1** and the plurality of second conductive terminals

52 having the second ground terminal G2. The grounding piece 6 includes the first grounding tab 621 and the second grounding tab 622. By making the first grounding tab 621 and the second grounding tab 622 contact the first tail portion 422 of the first ground terminal G1 and the second tail portion 522 of the second ground terminal G2, respectively, it simplifies the structure of the grounding piece 6 and also enables a shielding function.

(48) As shown in FIG. 16 and FIG. 17, the electrical connector 100 is connected to the plurality of cables 200. In the illustrated embodiment of the present disclosure, the cable 200 is a coaxial cable which includes a plurality of inner conductors 204, a plurality of inner insulating layers 205, a shielding layer 206 wrapped around at least part of the inner insulating layers 205, and an outer insulating layer 207 wrapped around at least part of the shielding layer 206. The inner insulating layer 205 is wrapped around at least part of a corresponding inner conductor 204. The inner conductor 204 may be tin-plated copper or silver-plated copper.

(49) When the plurality of cables 200 are connected to the electrical connector 100, the inner conductors 204 of the cables 200 are fixedly connected to the first tail portions 422 of the corresponding first conductive terminals 42 and the second tail portions 522 of the corresponding second conductive terminals 52, through soldering or welding for example, to achieve electrical connection. The shielding layer 206 is connected to the grounding piece 6 through soldering or welding for example, thereby increasing the shielding area and improving the shielding effect. When the cables 200 adopts coaxial cables of the present disclosure, the cables 200 can be directly soldered or welded to the electrical connector 100, which simplifies the process and reduces the cost.

(50) The above embodiments are only used to illustrate the present disclosure and not to limit the technical solutions described in the present disclosure. The understanding of this specification should be based on those skilled in the art. Descriptions of directions, although they have been described in detail in the above-mentioned embodiments of the present disclosure, those skilled in the art should understand that modifications or equivalent substitutions can still be made to the application, and all technical solutions and improvements that do not depart from the spirit and scope of the application should be covered by the claims of the application.

Claims

1. An electrical connector, comprising: an insulating body, the insulating body comprising a mating surface, a mounting surface opposite the mating surface, a mating slot extending through the mating surface, and a mounting space extending through the mounting surface; the mounting space being in communication with the mating slot; a first terminal module, the first terminal module comprising a first insulating block and a plurality of first conductive terminals, each first conductive terminal comprising a first fixing portion fixed to the first insulating block, a first resilient arm extending from the first fixing portion, and a first tail portion; the first resilient arm being provided with a first contact portion extending into the mating slot; a second terminal module, the second terminal module comprising a second insulating block and a plurality of second conductive terminals, each second conductive terminal comprising a second fixing portion fixed to the second insulating block, a second resilient arm extending from the second fixing portion, and a second tail portion, the second resilient arm being provided with a second contact portion extending into the mating slot, the first contact portions of the first conductive terminals and the second contact portions of the second conductive terminals being located on opposite sides of the mating slot, respectively; and a grounding piece, the grounding piece comprising a body portion and a plurality of grounding tabs connected to the body portion, the plurality of grounding tabs comprising a first grounding tab and a second grounding tab; wherein the first tail portions of the plurality of first conductive terminals and the second tail portions of the plurality of second conductive terminals are located on a same side and arranged in a row in a first direction; the

plurality of first conductive terminals comprise a first ground terminal; the plurality of second conductive terminals comprise a second ground terminal; the first grounding tab is stacked at least partially on the first tail portion of the first ground terminal and is in contact with the first tail portion of the first ground terminal; and the second grounding tab is stacked at least partially on the second tail portion of the second ground terminal and is in contact with the second tail portion of the second ground terminal.

2. The electrical connector according to claim 1, wherein the first tail portion of the first ground terminal comprises a first plane, and the first grounding tab comprises a first abutting plane in contact with the first plane; and wherein the second tail portion of the second ground terminal comprises a second plane, and the second grounding tab comprises a second abutting plane in contact with the second plane.

3. The electrical connector according to claim 2, wherein the first grounding tab is fixed to the first tail portion of the first ground terminal by soldering or welding; and the second grounding tab is fixed to the second tail portion of the second ground terminal by soldering or welding.

4. The electrical connector according to claim 1, wherein the first grounding tab comprises a first root portion connected to the body portion, a first inclined portion connected to the first root portion, and a first extension portion connected to the first inclined portion, the first extension portion is in contact with the first tail portion of the first ground terminal; and wherein the second grounding tab comprises a second root portion connected to the body portion, a second inclined portion connected to the second root portion, and a second extension portion connected to the second inclined portion, the second extension portion is in contact with the second tail portion of the second ground terminal.

5. The electrical connector according to claim 1, wherein the grounding piece further comprises a first spring arm extending from one side of the body portion and a second spring arm extending from another side of the body portion, the first spring arm is provided with a first curved protrusion protruding away from the second spring arm, and the second spring arm is provided with a second curved protrusion protruding away from the first spring arm.

6. The electrical connector according to claim 5, further comprising a metal shell sleeved on the insulating body; wherein the first spring arm and the second spring arm are abutted against the metal shell.

7. The electrical connector according to claim 5, wherein the grounding piece comprises a first fixing tab extending from the first spring arm and a second fixing tab extending from the second spring arm; and wherein the first fixing tab and the second fixing tab are inserted in and fixed to the first insulating block.

8. The electrical connector according to claim 7, wherein the grounding piece further comprises a third fixing tab bent from the body portion, and the third fixing tab is inserted in and fixed to the first insulating block.

9. The electrical connector according to claim 1, wherein the grounding piece further comprises a back-up grounding tab extending from the body portion, and the first insulating block defines a recess to receive the back-up grounding tab.

10. The electrical connector according to claim 1, further comprising a plurality of cables, part of the plurality of cables are connected to part of the first tail portions and another part of the plurality of cables are connected to part of the second tail portions; wherein the plurality of cables comprise a ground cable connected to the body portion.

11. The electrical connector according to claim 10, wherein the cable comprises a plurality of inner conductors, a plurality of inner insulating layers, a shielding layer wrapped around at least part of the inner insulating layers, and an outer insulating layer wrapped around at least part of the shielding layer; each inner insulating layer is wrapped around a corresponding inner conductor; and wherein the inner conductors of the cable are fixedly connected to the first tail portion of a corresponding first conductive terminal and to the second tail portion of a corresponding second

conductive terminal; the shielding layer is connected to the grounding piece.

12. An electrical connector, comprising: an insulating body comprising a mating surface and a mating slot extending through the mating surface; a first terminal module comprising a plurality of first conductive terminals, each first conductive terminal comprising a first resilient arm and a first tail portion, the first resilient arm being provided with a first contact portion extending into the mating slot; a second terminal module comprising a plurality of second conductive terminals, each second conductive terminal comprising a second resilient arm and a second tail portion, the second resilient arm being provided with a second contact portion extending into the mating slot, the first contact portions of the first conductive terminals and the second contact portions of the second conductive terminals being located on opposite sides of the mating slot, respectively; a grounding piece comprising a body portion and a plurality of grounding tabs connected to the body portion, the plurality of grounding tabs comprising a first grounding tab and a second grounding tab; and a plurality of cables, part of the plurality of cables being connected to part of the first tail portions of the first conductive terminals, and another part of the plurality of cables are connected to part of the second tail portions of the second conductive terminals; the plurality of cables comprising a ground cable connected to the body portion; wherein the first tail portions of the plurality of first conductive terminals and the second tail portions of the plurality of second conductive terminals are located on a same side and arranged in a row in a first direction; the plurality of first conductive terminals comprise a first ground terminal; the plurality of second conductive terminals comprise a second ground terminal; the first grounding tab is stacked at least partially on the first tail portion of the first ground terminal and is in contact with the first tail portion of the first ground terminal; and the second grounding tab is stacked at least partially on the second tail portion of the second ground terminal and is in contact with the second tail portion of the second ground terminal.

13. The electrical connector according to claim 12, wherein the first tail portion of the first ground terminal comprises a first plane, and the first grounding tab comprises a first abutting plane in contact with the first plane; and wherein the second tail portion of the second ground terminal comprises a second plane, and the second grounding tab comprises a second abutting plane in contact with the second plane.

14. The electrical connector according to claim 13, wherein the first grounding tab is fixed to the first tail portion of the first ground terminal by soldering or welding; and the second grounding tab is fixed to the second tail portion of the second ground terminal by soldering or welding.

15. The electrical connector according to claim 12, wherein the first grounding tab comprises a first root portion connected to the body portion, a first inclined portion connected to the first root portion, and a first extension portion connected to the first inclined portion, the first extension portion is in contact with the first tail portion of the first ground terminal; and wherein the second grounding tab comprises a second root portion connected to the body portion, a second inclined portion connected to the second root portion, and a second extension portion connected to the second inclined portion, the second extension portion is in contact with the second tail portion of the second ground terminal.

16. The electrical connector according to claim 12, wherein the grounding piece further comprises a first spring arm extending from one side of the body portion and a second spring arm extending from another side of the body portion, the first spring arm is provided with a first curved protrusion protruding away from the second spring arm, and the second spring arm is provided with a second curved protrusion protruding away from the first spring arm.

17. The electrical connector according to claim 16, further comprising a metal shell sleeved on the insulating body; wherein the first spring arm and the second spring arm are abutted against the metal shell.

18. The electrical connector according to claim 16, wherein the grounding piece comprises a first fixing tab extending from the first spring arm and a second fixing tab extending from the second spring arm; and wherein the first fixing tab and the second fixing tab are inserted in and fixed to the

first insulating block.

19. The electrical connector according to claim 18, wherein the grounding piece further comprises a third fixing tab bent from the body portion, and the third fixing tab is inserted in and fixed to the first insulating block.

20. The electrical connector according to claim 18, wherein the grounding piece further comprises a back-up grounding tab extending from the body portion, and the first insulating block defines a recess to receive the back-up grounding tab.
